

Inversive Geometry

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1 Inversion

We are quite familiar with some of the plane transformations at this point. This includes central and axial symmetry, rotation, and homothety – all of them translate lines into lines and circles – into circles. Neither symmetry nor rotation, nor homothety can transform a line into a circle or, conversely, a circle into a straight line since from the point of view of these transformations, a straight line and a circle are different objects. We call such transformations *linear transformations*.

There is, however, a remarkable transformation, the inversion, which translates a straight line into a straight line or a circle, and a circle into a circle or a straight line. Thus, from the point of view of inversion, a line and a circle turn out to be the same objects.

Definition 1. Consider a circle ω with a center O and radius r . The **inverse** M^* of a point $M \neq O$ with respect to ω (or *with center at O and degree r^2*) is a point on ray OM such that $OM \cdot OM^* = r^2$.

Inversion with the center O is a transformation of the plane with the point O gouged out (since the image of the point O is not defined). Nevertheless, we will take the liberty of speaking and speak of the image of a straight line or a circle passing through O .

The definition gives us a few properties right away:

1. For each point of the plane, except point O , there exists a unique inverse w.r.t. circle ω .
2. Every point on the circle ω translates into itself.
3. If a point M lies inside ω , then M^* lies outside ω , and vice versa.
4. (Involution) If M^* is the inverse of M , then M is the inverse of M^* .

Note 1. X translates into Y during an inversion = X inverts to Y .

Now we can consider a transformation (of the whole plane onto itself) that inverts every point, except the center O , into its inverse with respect to a circle ω . This is exactly what we call the **inversion with respect to ω** .

Problem 1. Consider a point $M(x, y) \in \mathbb{R}$ and a circle $\omega : x^2 + y^2 = r^2$ for some $r > 0$. Find the coordinates of M^* which is defined as the inverse of M w.r.t. circle ω .

Exercise 1. Consider a point A which lies outside of a given circle ω with center O and radius r , tangents AM and AN from A to ω , and point $B = OA \cap MN$. Show that points A and B are inverses of each other.

Problem 2. Find another way to construct an inverse of a point with respect to a given circle.

Hint: it can be done with only 3 lines and one circle.

So far we have only been inverting points, but let's take a look at what happens if we try to invert more complex objects instead. Naturally, lines come to mind first! If this line goes through the center of our inversion, it is pretty clear that the line inverts onto itself (Why?). Much more interesting case is when the line does not go through the center. Before investigating this case deeper, let's prove the following fact:

Exercise 2. Consider pairs A, A^* , and B, B^* of different points and their inverses with respect to a circle ω with center O . Show that $\angle OAB = \angle OB^*A^*$.

Exercise 3. Calculate the length of A^*B^* given the side lengths of $\triangle OAB$ and the radius of inversion r .

Problem 3. Points A and B lie on a circle ω . Tangents to ω passing through A and B intersect at P . Prove that P is the inverse of the middle point of AB considering an inversion w.r.t. ω . (Does this remind you of anything?)

Theorem 1. A line that does not go through the center of inversion inverts into a circle that goes through the center of inversion.

Note 2. Don't forget about the vice versa direction: a circle that goes through the center of inversion inverts into a line that does not go through the center of inversion.

Exercise 4. Let circle α be the image of a line a after an inversion w.r.t. circle ω . Consider a point A on the line a . Prove that the tangents from A to circles α and ω are equal length.

Theorem 2. A circle not going through the center of inversion inverts onto a circle that also does not go through the center of inversion.

Note 3. One important remark is that tangent lines/circles (so lines/circles that intersect only once, including the point of infinity for two lines) remain tangent under inversion. This has the power to send tangent circles to parallel lines—we simply invert around the point at which they are internally or externally tangent.

Theorem 3. Let O be the center of homothety between the circles α_1 and α_2 with point $A_1 \in \alpha_1$ is mapped to point $A_2 \in \alpha_2$. Let line A_1A_2 intersect α_1 and α_2 in points B_1 and B_2 , respectively. Then the product $OA_1 \cdot OB_2$ is constant and does not depend on the choice of the point A_1 .

Exercise 5. Let ABC be a triangle with orthocenter H and altitudes AD , BE , and CF . Where do the six points go if we invert everything around C with radius $\sqrt{CH \cdot CF}$?

Exercise 6. What does the circumcircle of $\triangle ABC$ invert into w.r.t. the incircle of $\triangle ABC$?

Exercise 7. Consider altitudes AA_1 and BB_1 in an acute triangle ABC .

1. What is the inverse of the line A_1B_1 during the inversion with center C that inverts A_1 into B ?
2. Line A_1B_1 intersects circumcircle ABC in points X and Y . Circumcircle around CBB_1 intersects the altitude AA_1 in point Z . Prove that $CX = CY = CZ$.

1.1 Cookie #1.

Now, there are also some specific tricks with inversion that come in handy once in a blue moon. First one is called $\sqrt{bc}$ **inversion**.

Problem 4. Let ABC be a triangle where X is the A -excenter and I is the incenter. Let M be the circumcenter of $\triangle BIC$, and let G be the point on BC such that $XG \perp BC$. Construct a circle Ω with diameter AX . If Ω and the circumcircle of $\triangle ABC$ intersect at a second point P , prove that M , G , and P are collinear.

Note 4. A -excenter is the center of an excircle tangent to a , which is the opposite to A .

Definition 2. $\sqrt{bc}$ inversion is the inversion about A with radius $\sqrt{AB \cdot AC}$ in $\triangle ABC$ and reflecting the resulting image over the A -angle bisector. In particular, this transformation swaps B and C , so that $\triangle ABC$ is fixed.

Note 5. In Problem 4 the reflection was indeed not needed since $MB = MC$.

Exercise 8. In triangle ABC with circumcircle Γ , the angle bisector of $\angle A$ intersects BC at D and Γ again at E . The circle with diameter DE meets Γ again at F . Prove that AF is a symmedian of triangle ABC .

Problem 5 (Japan, 2009). Triangle ABC has circumcircle Γ . A circle with center O is tangent to BC at P and internally to Γ at Q , so that Q lies on arc BC of Γ not containing A . Prove that if $\angle BAO = \angle CAO$ then $\angle PAO = \angle QAO$.

1.2 Cookie #2.

So, during inversion circles translate into circles. Let us define the angle between two intersecting circles as the angle between the tangent lines at the point of intersection point of the circles.

Theorem 4. Angle between two intersecting circles stays the same during an inversion.

Problem 6. Let ω_1 and ω_2 be two circles centered at O and P , and suppose they intersect at two points X and Y such that $\angle OXP = \angle OYP = 90^\circ$. Diameter AB of ω_1 is selected so that B lies strictly inside ω_2 . The two circles tangent to ω_2 , passing through O and A , touch ω_2 at F and G . Prove that $FGOB$ is cyclic.

In the problem above we encountered **orthogonal circles**, circles that have their intersection points form right angles with circles' centers. Such circles remain fixed when inverted about the other.

Exercise 9. Prove that orthogonal circles stay fixed during an inversion w.r.t. each of the circles, but the individual points do not.

Exercise 10. Consider points A_1 and A_2 that are symmetric w.r.t. circle ω . Show that an arbitrary circle α passing through A_1 and A_2 is orthogonal to ω .

Problem 7. Construct a circle that goes through a given point and orthogonal to two given circles.

Problem 8. Consider a pair of orthogonal circles ω_1 and ω_2 which intersect in points M and N . Let AB be the diameter of ω_1 . Prove that lines AM , AN , BM , and BN intersect ω_2 in the end points of a diameter that is orthogonal to AB .

Problem 9. Let the altitudes AM and BN of $\triangle ABC$ intersect in point D , Then the circumcircles of quadrilaterals $ABMN$ and $CMDN$ are orthogonal.

1.3 Cookie #3.

Inversion pops up a lot in the Japanese temple geometry, but it got a bit of a twist to it!

Note 6. In the next few problems, a circle called ω_i has center O_i and radius r_i unless specified otherwise.

Exercise 11. Through point P inside a circle ω four circles $\omega_1, \omega_2, \omega_3, \omega_4$, all internally tangent to ω were drawn. Moreover, pairs of circles ω_1 and ω_3, ω_2 and ω_4 are tangent at point P . Prove that $\frac{1}{r_1} + \frac{1}{r_3} = \frac{1}{r_2} + \frac{1}{r_4}$.

Exercise 12. Rays TX and TY intersect a circle ω with radius r and center O in four points. Four circles $\omega_1, \omega_2, \omega_3, \omega_4$ are all tangent to TX, TY , and ω in a way that $r_i < r_{i+1}$ for $i = 1, 2, 3$. Prove that $r_1 r_4 = r_2 r_3$.

Problem 10. Circles ω_1 and ω_2 are externally tangent to each other and a line l . Three circles $\omega_3, \omega_4, \omega_5$ are externally tangent to each other while ω_3 and ω_4 are also tangent to ω_1, ω_3 and ω_5 are also tangent to ω_2 , and ω_4 and ω_5 are also tangent to l . Let d denote the distance from O_3 to l . Find r_3 with respect to r_1 and r_2 , and prove that $d = 3r_3$.

Now, let's add a few more circles and bring some other techniques in the room!

Problem 11. Circles ω_1 and ω_2 with equal radii r are externally tangent to each other and a line l . A circle ω_3 is externally tangent to ω_1, ω_2 , and the line l , a circle ω_4 is externally tangent to $\omega_1, \omega_2, \omega_3$, a circle ω_5 is externally tangent to $\omega_1, \omega_2, \omega_4$, and so on. Find r_n for $n = 3, 4, \dots$ with respect to r .

What is going to be different if the initial circles ω_1 and ω_2 do not have equal radii?

Problem 12. Consider a circle ω with center at C and radius r inscribed in a semicircle Γ with radius $2r$ and center O which is a midpoint of diameter AB . Consider a circle ω_1 that is tangent to AB, Γ , and ω , a circle ω_2 that is tangent to ω_1, Γ, ω , a circle ω_3 that is tangent to ω_2, Γ, ω , and so on. Find r_n for $n = 3, 4, \dots$ with respect to r .

1.4 Problems for Inversion

Problem 13. Consider the altitude AH in an acute triangle ABC together with the projections K and L of H onto sides AB and AC . Circumcircle Ω of $\triangle ABC$ intersects the line KL in points P and Q , and the line AH – in points A and T . Prove that H is the center of the incircle of $\triangle PQT$.

Problem 14. Two circles intersect in points A and B . The common tangent of the circles (that is closer to the point B) is tangent to the circles at points E and F . The line AB intersects the line EF in point M . A point K is chosen on the extension of AM beyond point M such that $KM = MA$. Line KE intersects the circle containing point E for the second time at point C . Similarly, line KF intersects the circle containing point F for the second time at point D . Prove that points A , C , and D are collinear.

Problem 15. Consider a circle and a point P , distinct from the center, inside of it. Consider pairs of circles tangent to the given circle from the inside and to each other at the point P . Find the locus of intersection points of common external tangents to these circles.

Problem 16. Consider a quadrilateral $ABCD$ with its inscribed circle having center I . Tangents to the circumcircle of $\triangle AIC$ at points A and C intersect in point X . Tangents to the circumcircle of $\triangle BID$ at points B and D intersect in point Y . Prove that points X , I , Y are collinear.

Problem 17. A circle ω is inscribed in a quadrilateral $ABCD$ such that it is tangent to its sides BC and DA in points E and F , respectively. Lines AB , FE , and CD meet at one point S . Circumcircles Ω and Ω_1 of $\triangle AED$ and $\triangle BFC$ intersect ω for the second time in points E_1 and F_1 . Prove that lines EF and E_1F_1 are parallel.

Problem 18. Legs AB and CD of a trapezoid $ABCD$ are chords of respective circles ω_1 and ω_2 which are externally tangent. Let the degree measures of arcs AB and CD be α and β , respectively. Circles ω_3 and ω_4 also have chords AB and CD , respectively. Their arcs AB and CD , located on the same side of the chords as the corresponding arcs of the first two circles, have degree measures β and α . Prove that ω_3 and ω_4 are also tangent.

Problem 19. Angle AOB can be rotated into COD such that the ray OA becomes ray OC and ray OB becomes ray OD . There are circles inscribed into both of the angles such that the two circles intersect in points E and F . Prove that the angles AOE and DOF are equal.

Problem 20. Let $ABCD$ be a quadrilateral whose diagonals AC and BD are perpendicular and intersect at E . Prove that the reflections of E across AB , BC , CD , and DA are concyclic.

Problem 21. Quadrilateral $ABCD$ is inscribed into a circle Ω with center in O with O not lying on any of the diagonals of the quadrilateral. The middle point of the diagonal BD lies on the circumcircle Ω_1 of the triangle AOC . Prove that the middle point of the diagonal AC lies on a circumcircle Ω_2 of the triangle BOD .

Problem 22. The orthocenter H of a triangle ABC lies on its incircle. Prove that the three circles with centers at A , B , and C , respectively, all passing through the point H , have a shared tangent line.

2 Poles and Polars

Let us note that there are exactly two points M and N on a circle that translate to themselves during a projection of a circle onto itself with an outside center. To construct those, one just needs to construct tangent lines from the center of the projection.

Now, let us consider an arbitrary chord AB with the respective image points A' and B' . By the Butterfly theorem we know that $A'B$ and AB' intersect MN at the same point. We actually know more: AB and $A'B'$ intersect MN at the same point. This, by the way, gives a very nice way to construct the tangent line from a point to a circle with only a ruler. However, there is another way to describe the line MN .

Definition 3. Consider a circle ω with center O . The line p perpendicular to the ray OP at P^* is called the **polar** of point P w.r.t. ω . Moreover, P is called the **pole** of line p w.r.t. ω .

Definition 4 (Polar Correspondence). A **polar correspondence** with respect to a circle ω is a mapping π which maps every point to its *polar*, and each line, which does not go through the center of ω , its pole. This mapping is an incidence relation:

$$\text{point } A \text{ lies on a line } l \Leftrightarrow \text{point } \pi(l) \text{ lies on the line } \pi(A).$$

Before moving on, answer the following questions:

- If B is an arbitrary point on the polar, what are the possible values of $\angle POB$?
- Can we construct the pole from the polar?
- Can a polar go through a pole?
- What can we say about a polar if its pole is outside of the circle?

Note that point O does not have a polar and a line through O does not have any poles on it.

Exercise 13. Find two ways to construct the polar of a point.

Exercise 14 (La Hire's Theorem?). If a point X lies on the polar of point Y w.r.t. ω , then the point Y lies on the polar of point X w.r.t. ω .

Exercise 15. Consider points A , B , and C with a , b , and c being their polars w.r.t. circle ω . Prove that the points A , B , and C are collinear if and only if lines a , b , and c are concurrent.

Theorem 5 (Brokard's Theorem). The points A , B , C , D lie in this order on circle ω with center O . AC and BD intersect at P , AB and DC intersect at Q , AD and BC intersect at R . Then O is the orthocenter of $\triangle PQR$. Furthermore, QR is the polar of P , PQ is the polar of R , and PR is the polar of Q w.r.t. ω .

Convince yourself in the following table based on the *Principle of duality*:

point	line
pole	polar
lies on	goes through
concurrent	collinear
3-vertex figure (triangle)	3-sided figure (triangle)
4-vertex figure (quadrilateral, 4 points and 6 lines)	4-sided figure (figure, 4 lines and 6 intersection points)
tangent line	tangent point
pole of a secant $AB \neq \text{diameter}$	intersection point of tangents at points A and B
polar of an outside point P	line connecting tangency points of the tangent lines from P
pole of a line $p \neq \text{diameter}$	intersection point of polars of two outside points on p
polar of a point $P \neq O$	line that connects poles of two secants that go through P

Exercise 16. Translate Brianchon's theorem into Pascal's theorem using the principle of duality.

Note 7. Recall the following:

- *Brianchon's theorem: principal diagonals of an circumscribed hexagon (those connecting opposite vertices) meet in a single point.*
- *Pascal's theorem: points of intersection of opposite sides of an circumscribed hexagon are collinear.*

Exercise 17. A triangle ABC is inscribed in a circle. Tangent lines in vertices A , B , and C intersect the lines BC , AC , and AB in points A_1 , B_1 , and C_1 , respectively. Prove that A_1 , B_1 , and C_1 are concurrent.

Exercise 18 (Harmonic quadrilateral). Consider an inscribed in a circle ω quadrilateral $ABCD$. It is known that the tangents to ω through points A and C either intersect on BD or parallel to BD . Prove that the tangents to ω through points B and D , either intersect on AC or parallel to it.

Exercise 19 (Inscribed quadrilateral). A quadrilateral $ABCD$ is inscribed in a circle with center O . Diagonals of $ABCD$ intersect at point P , lines AB and CD - at point R , and lines BC and DA - at point Q . Prove that points O , P , Q , and R are orthocentric.

*Note 8. Four points are called **orthocentric** when each point is the orthocentre of the triangle defined by the other three points.*

Exercise 20. Let A , B , C , and D be points on a circle in that order, and let P be the intersection of AB and CD . Let PE and PF be tangents to the circle. Prove that AC , BD , and EF are concurrent.

Exercise 21. Let UV be a diameter of a semicircle, and let P , Q are two points on the semicircle. The tangents to the semicircle at P and Q meet at R , and lines UP and VQ meet at S . Prove that $RS \perp UV$.

2.1 Poles and Polars Problems

Problem 23. Let ABC be a triangle and I be its incenter. Let the incircle of ABC touch sides BC , CA , AB at D , E , F respectively. Let S denote the intersection of lines EF and BC . Prove that $SI \perp AD$.

Problem 24. A circle is inscribed in quadrilateral $ABCD$ so that it touches sides AB , BC , CD , DA at E , F , G , H respectively.

1. Show that lines AC , EF , GH are concurrent. In fact, they concur at the pole of BD .
2. Show that lines AC , BD , EG , FH are concurrent.

Problem 25. Fix a chord MN in a circle. For each diameter AB of the circle consider the intersection point of lines AM and BN , and draw a line l , which is also perpendicular to AB through it. Prove that all lines l meet at a single point.

Problem 26. Suppose $ABCD$ is a cyclic quadrilateral inscribed in the circle ω . Let $E = AB \cap CD$ and $F = AD \cap BC$. Let ω_1 and ω_2 be the circumcircles of triangles AEF and CEF , respectively. Let G and H be the intersections of ω and ω_1 , ω and ω_2 , respectively, with $G \neq A$ and $H \neq C$. Show that AC , BD , and GH are concurrent.

Problem 27 (China MO, 1996). Let H be the orthocenter of triangle ABC . From A construct tangents AP and AQ to the circle with diameter BC , where P , Q are the points of tangency. Prove that P , H , Q are collinear.

Problem 28. Let ABC be a triangle. The incircle touches the side BC at A' and the line AA' meets the incircle again at a point P . Let the lines CP and BP meet the incircle of triangle ABC again at N and M , respectively. Prove that the lines AA' , BN , and CM are concurrent.

Problem 29 (IMO 1985). A circle with center O passes through the vertices A and C of triangle ABC and intersects segments AB and BC again at distinct points K and N , respectively. The circumcircles of triangles ABC and KBN intersect at exactly two distinct points B and M . Prove that $\angle OMB = 90^\circ$.

Problem 30 (Brazil MO, final round, 2013). Consider an acute triangle ABC with altitudes AD , BE , CF . Line BE intersects the circumcircle of $\triangle ADC$ in points K and L (K lies between points B and L), and line CF intersects the circumcircle of $\triangle ADB$ in points P and Q (P lies between points C and Q). Prove that the intersection point of lines LP and KQ lies on BC .

Problem 31 (Vietnam MO, final round, 2012). Consider a quadrilateral $ABCD$ inscribed in a circle with center O . Rays AB and DC intersect at point M , and rays BC and AD intersect at point N . Let P , Q , S , and T be different points of intersection of bisectors of the following pairs of angles: $\angle MAN$ and $\angle MBN$, $\angle MBN$ and $\angle MCN$, $\angle MDN$ and $\angle MAN$, and $\angle MCN$ and $\angle MDN$, respectively.

- Prove that points P , Q , S , T are concyclic.
- Prove that points O , E , I are collinear, where E is the point of intersection of AC and BD .

3 Mix and Mash

Problem 32 (Russia MO, final round, 2013). Consider a triangle ABC with the inscribed circle ω with center I . Let Γ be the circumcircle around $\triangle AIB$. Circles ω and Γ intersect in points X and Y . Common tangents to circles ω and Γ intersect in point Z . Prove that the circumcircles around $\triangle ABC$ and $\triangle XYZ$ are tangent.

Problem 33 (India MO, final round, 2012). Quadrilateral $ABCD$ is inscribed in a circle with center O . Circumcircles of $\triangle AOB$ and $\triangle COD$ intersect for the second time at point P that lies inside $\triangle AOD$. Consider point Q on the ray OP past point P and point R on the ray PO past point O such that $\angle QAP = \angle OBR$. Prove that $\angle PDQ = \angle RCO$.

Problem 34 (USAMO, 2009). Trapezoid $ABCD$, with $AB \parallel CD$, is inscribed in circle ω and point G lies inside triangle BCD . Rays AG and BG meet ω again at points P and Q , respectively. Let the line through G parallel to AB intersect BD and BC at points R and S , respectively. Prove that quadrilateral $PQRS$ is cyclic if and only if BG bisects $\angle CBD$.

Problem 35 (Ukraine IMO TST 2023). Inscribed in $\triangle ABC$ circle with center in I is tangent to side BC at point D . Line that goes through A intersects lines BC , BI , CI in points X , Y , Z , respectively. Circumcircle of $\triangle ABC$ intersects circumcircles of $\triangle AIY$ and $\triangle AIZ$ for the second time in points U and V , respectively. Prove that points D , U , V , and X are collinear.

Problem 36 (US IMO TST 2012). In scalene triangle ABC , let the feet of the perpendiculars from A to BC , B to CA , C to AB be A_1, B_1, C_1 , respectively. Denote by A_2 the intersection of lines BC and B_1C_1 . Define B_2 and C_2 analogously. Let D, E, F be the respective midpoints of sides BC, CA, AB . Show that the perpendiculars from D to AA_2 , E to BB_2 and F to CC_2 are concurrent.

Problem 37 (IMO Shortlist 2003). Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1 and Γ_3 are externally tangent at P , and Γ_2 and Γ_4 are externally tangent at the same point P . Suppose that Γ_1 and Γ_2 , Γ_2 and Γ_3 , Γ_3 and Γ_4 , and Γ_4 and Γ_1 meet at A, B, C , and D , respectively, and that all these points are different from P . Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$

Problem 38 (IMO Shortlist 2019). Let I be the incentre of acute triangle ABC . Let the incircle meet BC, CA , and AB at D, E , and F , respectively. Let line EF intersect the circumcircle of the triangle at P and Q , such that F lies between E and P . Prove that $\angle DPA + \angle AQD = \angle QIP$.

Problem 39 (IMO Shortlist 2021). A point D is chosen inside an acute triangle ABC with $AB > AC$ so that $\angle BAD = \angle DAC$. A point E is constructed on the segment AC so that $\angle ADE = \angle DCB$. Similarly, a point F is constructed on the segment AB so that $\angle ADF = \angle DBC$. A point X is chosen on the line AC so that $CX = BX$. Let O_1 and O_2 be the circumcentres of the triangles ADC and DXE . Prove that the lines BC, EF , and O_1O_2 are concurrent.

Problem 40 (IMO Shortlist 2022). Let ABC and $A_1B_1C_1$ be two triangles having the same circumcircle ω , and the same orthocentre H . Let Ω be the circumcircle of the triangle determined by the lines AA_1, BB_1 , and CC_1 . Prove that H , the centre of ω , and the centre of Ω are collinear.